TrackRock

Institutional Home Acquisition Intelligence Platform

Project Technical Summary

Hackathon Submission — Real Estate Data Ingestion Engine

April 2026

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01 — EXECUTIVE SUMMARY

TrackRock is a web-based geospatial intelligence platform built to expose and analyze the large-scale acquisition of single-family homes by institutional investors — primarily BlackRock and its network of subsidiaries and affiliated LLCs. The tool is designed for investigative journalists, housing policy activists, academic researchers, and community organizers who need hard evidence of how concentrated institutional buying reshapes housing costs, community demographics, and neighborhood stability.

Unlike raw data dumps or generic real estate dashboards, TrackRock tells a story — layering ownership data against rent inflation, eviction rates, demographic shift, and home price trajectories to produce a living picture of institutional displacement. The platform combines a robust data ingestion pipeline, an entity resolution engine for tracing LLC subsidiaries back to their parent, and a Google Maps-powered geospatial interface that lets users explore submarkets at the city, zip code, and Census tract level.

The Core Question TrackRock Answers:

When BlackRock and its subsidiaries buy hundreds of homes in the same neighborhood, what actually happens to the people who already live there — and can we prove it with data?

02 — PRODUCT VISION & USER STORIES

Target Users

Investigative Journalist	Needs to identify acquisition clusters, pull a one-page evidence brief for a story, and map LLC chains back to BlackRock.
Housing Activist	Wants to show their community a visual map of who owns their block and how rents have changed since acquisitions began.
Policy Researcher	Needs tract-level data exports correlating ownership concentration with eviction and demographic data for academic papers.
City Council Staff	Needs a quick read on which neighborhoods in their district have the highest institutional exposure.

User Journey

The intended experience flows in three acts that mirror the narrative arc of the story TrackRock tells:

- Act 1 — WHERE DID THEY BUY?
  - User lands on the map. A heatmap of acquisition density is immediately visible across the selected city.
  - A timeline scrubber lets them drag through years, watching the cluster grow in real time.
  - Filters let them narrow by acquisition year, entity name, or concentration score.

- Act 2 — WHAT HAPPENED NEXT?
  - User toggles overlay layers: rent index change, home price index, eviction rate, permit activity.
  - Each layer tells the next chapter — did rents spike after a cluster formed? Did evictions rise?
- Act 3 — WHO LIVES THERE NOW?
  - User drills into a Census tract to see demographic composition, homeownership rate change, and median income shifts over time.
  - A 'Generate Report' button produces a one-page PDF brief with all key metrics for that tract.

03 — TECHNOLOGY STACK

Frontend

Framework	React 18 with TypeScript — component-based UI, strongly typed for reliability at scale.
Mapping	Google Maps JavaScript API — primary map interface with heatmap, polygon, and marker overlay layers.
Styling	Tailwind CSS — utility-first styling for rapid UI development. Dark, high-contrast investigative aesthetic.
State Management	Zustand — lightweight global state for active filters, selected tract, and layer visibility.
Charting	Recharts — time-series charts for rent index, price index, and eviction trends displayed in the sidebar panel.
PDF Export	react-pdf / @react-pdf/renderer — generates the one-page Neighborhood Report brief on demand.

Backend

Runtime	Node.js with Express — REST API server handling data requests, pipeline triggers, and Gemini AI calls.
Database	PostgreSQL with PostGIS extension — stores normalized property, tract metric, and ownership data with full geospatial query support.
ORM	Prisma — type-safe database access layer, schema migrations.
Geocoding	Google Maps Geocoding API — converts raw property addresses to lat/lng coordinates and FIPS tract codes.
Caching	Redis — caches expensive aggregation queries (concentration scores, tract rollups) for sub-second map tile loads.
Task Queue	BullMQ — manages background pipeline jobs for data ingestion and refresh cycles.

AI Layer

Model	Google Gemini 1.5 Pro via Gemini API — powers the Narrative Intelligence Engine (see Section 6).

Use Cases	LLC entity resolution from ambiguous owner names; natural language neighborhood report generation; query assistant for non-technical users.
Integration	Called server-side via the @google/generative-ai SDK. Responses are cached and structured into JSON before being returned to the frontend.

Infrastructure & DevOps

Hosting	Google Cloud Platform (GCP) — Cloud Run for the backend API container, Cloud SQL (PostgreSQL) for managed database.
Storage	Google Cloud Storage — raw CSV uploads, pipeline artifacts, and generated PDF reports.
CI/CD	GitHub Actions — automated test, build, and deploy pipeline to Cloud Run on merge to main.
Containerization	Docker — consistent local dev and production environments.
Environment	dotenv — API key and secrets management for local development.

04 — DATA SOURCES & APIs

Primary Proprietary Data

The foundational dataset is the BlackRock property acquisition CSV provided as input, with columns: Property_ID, Owner_Name, Mailing_Address, Property_Address, and Match_Reason. This is the spine of the platform — every other data source annotates or contextualizes it.

Critical Note on Owner_Name:

BlackRock does not appear by name on most deeds. Properties are held through LLCs such as 'IH4 PROPERTY ILLINOIS LLC', 'SFR JV-1 PROPERTY LLC', and hundreds of others. The entity resolution pipeline (Section 5) is what maps these back to the parent entity.

External Data APIs & Sources

API / Source	Provider	Data Provided	Cost
Google Maps Geocoding API	Google	Converts property addresses to lat/lng + Census tract FIPS codes. Essential spine of all geospatial joins.	Paid (free tier: 40k/mo)
Google Maps JavaScript API	Google	Primary map rendering engine. Heatmap layer, polygon overlays, custom markers, info windows.	Paid (free tier included)
Gemini 1.5 Pro API	Google	LLC entity resolution, narrative report generation, natural language query interface.	Paid (low cost per token)

Census ACS API	U.S. Census Bureau	Median household income, race/ethnicity, homeownership rate, renter burden — by Census tract.	Free
FRED API	St. Louis Fed	Metro-level rent index, housing price index, vacancy rates, employment growth time series.	Free
HUD Fair Market Rents	HUD	Annual fair market rents by metro area. Used to benchmark whether rents in high-concentration tracts exceed FMR.	Free CSV
Zillow Research Data	Zillow	ZHVI (home value index) and ZORI (rent index) by zip code over time. Key source for price change visualization.	Free CSV download
Eviction Lab API	Princeton University	Eviction filing rates and eviction rates by Census tract. The 'human cost' layer of the platform.	Free (registration req.)
OpenCorporates API	OpenCorporates	LLC registration lookup — helps trace subsidiary chains, registered agents, and incorporation states.	Freemium
Local Building Permit APIs	City Open Data Portals	New construction permit counts by tract. Shows whether supply is growing in high-concentration areas.	Free (varies by city)

05 — DATA PIPELINE ARCHITECTURE

Pipeline Overview

The data pipeline is organized into four sequential stages. Each stage produces a clean output that feeds the next, and all stages are orchestrated by BullMQ job queues running on the backend server.

Stage 1 — Ingestion

Raw data is pulled from all configured sources and stored as-is in Google Cloud Storage. No transformation occurs at this stage.

- BlackRock CSV is uploaded via the admin interface and stored in GCS.
- Census ACS, FRED, HUD, Zillow CSVs are fetched via API or direct download on a configurable schedule (weekly refresh).
- Eviction Lab and OpenCorporates data is fetched via REST API call.
- Local permit data is fetched from city open data portals (Chicago, Atlanta, Phoenix, Las Vegas — initial markets).

Stage 2 — Entity Resolution (The LLC Mapper)

This is the most technically novel component of the pipeline. The Owner_Name field in the source CSV contains hundreds of LLC names that must be mapped back to BlackRock or its known subsidiaries.

- A curated seed list of ~200 known BlackRock subsidiary names and LLC patterns (sourced from investigative journalism by ProPublica, The Atlantic, Bloomberg) is maintained in the database.
- Fuzzy string matching (using the Jaro-Winkler algorithm) catches typos and minor variations in LLC names.

- For names that don't match the seed list, Gemini 1.5 Pro is called with the owner name, mailing address, and match reason as context, and asked to classify whether the entity is likely affiliated with BlackRock and which subsidiary chain it belongs to.
- OpenCorporates API is queried for the LLC's registered agent — if the registered agent matches known BlackRock law firms or agent addresses, confidence score increases.
- All matches are stored with a confidence score (0–100) and the reasoning trace, so users can inspect why a property was attributed to BlackRock.

Stage 3 — Geocoding & Tract Assignment

Every property address is geocoded and assigned to the appropriate Census geography using the Google Maps Geocoding API.

- Google Maps Geocoding API converts Property_Address to lat/lng coordinates.
- Reverse geocoding and Census Geocoder API assign each property to its FIPS tract code.
- The FIPS code is the universal join key — it links property ownership data to ACS income data, Eviction Lab data, Zillow zip-level data, and FRED metro-level data.
- Geocoding results are cached in PostgreSQL to avoid redundant API calls on pipeline re-runs.

Stage 4 — Normalization & Aggregation

Clean, standardized records are written to the PostgreSQL/PostGIS database. Aggregation queries compute the metrics that power the map overlays.

- Properties table: property_id, lat, lng, fips_tract, owner_name, parent_entity, confidence_score, acquisition_date, purchase_price (if available).
- Tract metrics table: fips_tract, year, median_income, pct_renter, pct_white, pct_black, pct_hispanic, eviction_rate, permit_count.
- Price index table: zip_code, year, zhvi (home value), zori (rent index), fmr (fair market rent).
- Concentration score table: fips_tract, total_sfr_units, blackrock_owned_units, concentration_pct — computed quarterly.

Pipeline Refresh Schedule:

Property data: on upload (manual). Census ACS: annual (August refresh). Zillow indices: monthly. FRED series: monthly. Eviction Lab: annual. Permit data: weekly. Concentration scores: recomputed nightly.

06 — AI-POWERED FEATURES (GEMINI)

Feature 1 — LLC Entity Resolution

Described in Stage 2 above. Gemini 1.5 Pro is given the LLC name, mailing address, and any match metadata from the source CSV and asked to return a structured JSON object indicating parent entity classification, confidence, and reasoning. This replaces what would otherwise require a team of researchers manually tracing corporate registries.

Feature 2 — Neighborhood Report Narrative Generator

When a user clicks 'Generate Report' for a Census tract, the backend assembles all available metrics for that tract and passes them to Gemini with a structured prompt. Gemini returns a 3–4 paragraph plain-English narrative that becomes the opening section of the PDF report.

Example prompt context passed to Gemini:

Sample Gemini Prompt Structure:

"You are a housing policy analyst. Write a concise 3-paragraph summary for an investigative report about Census Tract 8501 in Fulton County, Atlanta. BlackRock-affiliated entities own 11.2% of single-family homes in this tract. Since 2015, median rent has increased 41% vs. a 22% national average. The eviction rate is 3.1x the county median. The homeownership rate has dropped from 54% to 38%. The tract is 73% Black. Write for an activist or journalist audience."

Feature 3 — Natural Language Query Assistant

A chat-style input in the sidebar allows non-technical users to query the dataset in plain English. Gemini translates the natural language query into a structured API call to the backend, which executes the appropriate database query and returns results.

- "Show me all tracts in Phoenix where BlackRock owns more than 5% of homes" → filter map to matching tracts.
- "Which neighborhood has seen the biggest rent increase after acquisitions?" → ranked list with map highlight.
- "Compare eviction rates in high-concentration vs. low-concentration tracts in Atlanta" → side-by-side chart.

07 — MAP INTERFACE & UI FEATURES

Google Maps Integration

The Google Maps JavaScript API is the primary rendering engine for all geospatial content. The map is the primary interface — not a feature within a dashboard, but the dashboard itself. All other UI elements (sidebar, filter panel, chart drawer) annotate the map.

Map Layers (Toggle-able Overlays)

Acquisition Heatmap	Density of BlackRock-affiliated properties by area. Default layer on load. Red intensity scales with concentration.
Concentration Score	Census tract polygons colored by % of SFR units owned by institutional investors. Choropleth scale from light to deep red.
Rent Change Overlay	Tract polygons colored by % rent increase since 2015 (Zillow ZORI). Toggle independently of ownership data.
Home Price Index	Zip-level ZHVI change shown as polygon shading. Allows visual correlation with acquisition clusters.
Eviction Rate Layer	Tract-level eviction rate from Eviction Lab. Purple scale — the 'human cost' layer.

Demographic Shift	Choropleth showing change in homeownership rate (2015 → present) per tract. Key equity indicator.
Permit Activity	Point markers for new building permit filings. Shows whether supply response is occurring.
Individual Properties	Dot markers for each BlackRock-attributed property. Click for property detail card.

Map Controls & Interactions

- Timeline Scrubber — horizontal slider beneath the map. Drag to animate acquisition spread by year (2012–present). Each tick re-renders the heatmap with only properties acquired through that year.
- Market Selector — dropdown to switch between supported cities (Atlanta, Phoenix, Las Vegas, Chicago). Triggers fresh pipeline pull and re-centers the map.
- Layer Toggle Panel — floating card (top right) with toggle switches for each data layer. Layers can be stacked.
- Filter Sidebar — left drawer with: year range slider, concentration score minimum, entity name search, neighborhood/zip search.
- Property Detail Card — clicking any dot marker opens a card with: address, attributed entity, confidence score, acquisition date, estimated current rent, nearby ZHVI trend sparkline.
- Tract Detail Panel — clicking any tract polygon opens the full tract metrics panel with tabbed charts (rent, eviction, demographics, permits) and the 'Generate Report' CTA.

Sidebar & Analytics Panel

- Market Summary Card — top-level stats for the current view: total properties tracked, number of unique entities, average concentration score, cities covered.
- Trend Charts (Recharts) — rent index over time, home price index, eviction rate trend. Linked to the currently selected tract or city.
- Entity Breakdown Table — ranked list of LLC names attributed to BlackRock in the current view, with property counts and confidence scores.
- Comparison Mode — select two tracts side-by-side to compare all metrics. Designed for the 'high concentration vs. low concentration' research use case.

Neighborhood Report (PDF Export)

The Neighborhood Report is the primary artifact TrackRock produces for activist and journalist users. It is generated on-demand for any selected Census tract and exported as a formatted PDF.

- Cover section: Tract name, county, city, date of report generation.
- AI Narrative (Gemini): 3–4 paragraph plain-English summary of the situation in the tract.
- Ownership Data: Number of BlackRock-attributed properties, concentration score, top 5 LLC names found, confidence score distribution.
- Economic Impact: Rent change %, home price change %, comparison to city and national averages.
- Human Cost: Eviction rate, comparison to county median, homeownership rate change.
- Demographic Context: Race/ethnicity composition, median income, renter vs. owner rate.
- Data Sources footnote and methodology note for journalistic credibility.

08 — COMPLETE FEATURE LIST

Core Features (Hackathon MVP)

- Google Maps heatmap of BlackRock property acquisitions by city.
- Timeline scrubber animating acquisition spread from 2012 to present.
- Eight toggle-able data overlay layers (ownership, rent, price, eviction, demographics, permits).
- Gemini-powered LLC entity resolution pipeline mapping subsidiaries to parent entity.
- Census tract detail panel with tabbed charts for all key metrics.
- On-demand Neighborhood Report PDF with Gemini-generated narrative.
- Natural language query assistant powered by Gemini 1.5 Pro.
- Multi-city support (Atlanta, Phoenix, Las Vegas, Chicago at launch).
- Confidence score transparency — every attribution shows its evidence.
- Data pipeline running on Google Cloud Platform with weekly refresh cycle.

Stretch / Bonus Features

- Email alert system — users subscribe to a tract and receive alerts when new acquisitions are detected.
- Corporate tree visualizer — interactive org chart showing the LLC subsidiary chain from property → LLC → parent entity.
- National leaderboard — which 10 census tracts in the U.S. have the highest BlackRock concentration score?
- 'Before & After' mode — side-by-side street view comparison using Google Maps Street View Static API.
- Data export (CSV/GeoJSON) for researchers who want to pull raw tract data into their own tools.
- Embeddable map widget for journalists to embed a live TrackRock map in their articles.

09 — PROJECT STRUCTURE & TIMELINE

Repository Structure

Monorepo Layout:

/trackrock /frontend → React + TypeScript app /src /components → Map, Sidebar, Charts, ReportModal /hooks → useMapLayers, useTractData, useGemini /stores → Zustand state (filters, selection, layers) /api → Frontend API client /backend → Node.js + Express API /pipeline → Ingestion, resolution, geocoding, normalization stages /routes → REST endpoints (tracts, properties, reports, query) /gemini → Gemini API integration wrappers /jobs → BullMQ job definitions /database /schema → Prisma schema + PostGIS setup /migrations → Database migration files /scripts /seed → Sample data loaders for demo /etl → Standalone pipeline scripts

Hackathon Build Timeline — Monday Apr 6 to Saturday Apr 11, 11am (~131 hours)

The schedule is organized into five phases across five working days. The extra time is used for polish, stretch features, and a clean demo — not scope creep.

Phase 1 — Foundation (Monday, ~8 hrs)

Mon AM	Monorepo scaffold (frontend + backend), GitHub repo, Docker setup. GCP project created, Cloud SQL (PostgreSQL + PostGIS) provisioned, Cloud Storage bucket configured.
Mon Midday	All API keys obtained and stored in .env: Google Maps Geocoding, Google Maps JS, Gemini 1.5 Pro. Prisma schema drafted, first migration run.
Mon PM	BlackRock CSV ingestion script written. Subsidiary seed list populated (~200 known LLC names from investigative journalism sources). BullMQ and Redis wired up.

Phase 2 — Data Pipeline (Tuesday, ~10 hrs)

Tue AM	Stage 2: Entity Resolution engine. Jaro-Winkler fuzzy matcher against seed list. Gemini LLC resolver for unmatched names — returns JSON with parent entity, confidence score, and reasoning.
Tue Midday	OpenCorporates API integration for registered agent lookup. Confidence scoring finalized and stored in DB.
Tue PM	Stage 3: Google Maps Geocoding API batch job — all property addresses geocoded to lat/lng + FIPS tract. Stage 4: Normalization jobs write to properties and tract_metrics tables.
Tue Eve	External data ingestion: Census ACS API, FRED API, Zillow ZHVI and ZORI CSV normalization. Concentration scores computed and stored.

Phase 3 — Backend API & Map Core (Wednesday, ~10 hrs)

Wed AM	Eviction Lab and HUD Fair Market Rent data ingested. City permit APIs (Atlanta, Phoenix) connected. All data tables seeded with demo data for four launch cities.
Wed Midday	REST API routes built: /tracts, /properties, /concentration, /metrics, /report. PostGIS spatial queries optimized. Redis caching layer added.
Wed PM	Frontend: Google Maps JS API integrated. Base map rendered. Heatmap layer from property lat/lng. Property dot markers with click handlers.
Wed Eve	Timeline scrubber wired to heatmap. Layer toggle panel (8 layers) scaffolded. Basic filter sidebar built.

Phase 4 — UI, AI Features & Report (Thursday + Friday, ~18 hrs)

Thu AM	Remaining map overlays: Concentration choropleth, Rent Change, Home Price, Eviction Rate, Demographic Shift, Permit Activity layers all wired to API.
Thu Midday	Tract detail panel: click polygon to open tabbed panel with Recharts time-series for rent, price, eviction, demographics. Entity breakdown table.
Thu PM	Gemini NL Query Assistant: sidebar chat input translates to API call, highlights results on map. Handles 10 common query types.
Thu Eve	Gemini Neighborhood Report narrative generator integrated. react-pdf report layout built with all metric sections.
Fri AM	PDF report polished end-to-end: cover, AI narrative, metric tables, data sources footnote. Generate Report button fully wired.
Fri Midday	Comparison Mode: select two tracts for side-by-side metric comparison. Corporate tree visualizer stretch feature attempted.
Fri PM	Full UI polish pass: dark investigative aesthetic, responsive layout, loading states, error handling, empty states.
Fri Eve	Multi-city support verified for Atlanta, Phoenix, Las Vegas, Chicago. Market selector dropdown wired. Demo data confirmed for all four cities.

Phase 5 — Deploy, Polish & Demo Prep (Saturday, 8am–11am)

Sat 8–9am	Deploy to Google Cloud Run. Smoke test all endpoints in production. Verify Google Maps API keys scoped correctly for deployed domain.
Sat 9–10am	Record 3-minute demo walkthrough video. Write submission README with architecture diagram, setup instructions, data source attribution.
Sat 10–11am	Final bug fixes only — no new features. Buffer time for last-minute deploy issues. Submit by 11am.

Time Allocation Philosophy:

With 131 hours available, the biggest risk is not running out of time — it is scope creep. Phases 1 through 3 are locked scope. Phase 4 is where extra time goes into polish rather than just functionality. Phase 5 is sacred: no new features after Friday midnight.

10 — THE NARRATIVE FRAMEWORK

TrackRock is designed around a single, unifying journalistic premise: institutional home buying is not an abstraction — it has names, addresses, and measurable consequences. Every design decision in this platform serves that premise.

The map is not a feature. It is the argument. When a user opens TrackRock and watches the heatmap bloom across a neighborhood between 2014 and 2019, they are not looking at data. They are watching a neighborhood change ownership — block by block, quarter by quarter — in a way that no family could compete with.

The Gemini-generated Neighborhood Report is not a convenience feature. It is a weapon for accountability journalism — a one-click dossier that any activist, council member, or reporter can print, cite, and hand to a legislator. The AI is not here to replace the journalist. It is here to do the work that would otherwise take three researchers two weeks, in thirty seconds.

> **The TrackRock Promise:**
> Open the app. Pick a city. In under 60 seconds, know more about who owns that neighborhood — and what happened because of it — than most local journalists who've covered that beat for years.

This is what makes TrackRock a unified product and not a bundle of features: every component — the pipeline, the entity resolver, the map layers, the AI report — serves the same user, answering the same question, in service of the same story.